higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

MATHEMATICS
(First Paper)
NQF LEVEL 4

21 FEBRUARY 2019

This marking guideline consists of 13 pages.

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INSTRUCTIONS TO THE MARKERS

1. Examiners are reminded that the subject Mathematics lends itself to a variety of problem solving methods and therefore several different alternative answers are possible. Allocate marks for ALL mathematically correct answers.

2. The answers provided in this memorandum are not exhaustive and examiners/markers are requested to discuss all additional alternative solutions before the commencement of marking. The marking memorandum strives to minimise discretionary marking to ensure consistency.

3. Implement "follow up" marking. In other words, allocate marks for accurate steps following an error, provided that the responses are consistent with the error. This is referred to as consistent accuracy (CA) marks. Be careful of double errors that result in an accurate answer.

4. Marks are allocated per step, but if a student omitted a step and there is evidence or reason to believe that he/she could derive the next step mentally, then the mark must still be given to the student.

5. Take note that marks have been allocated for simplification in certain questions. This is to identify high achievers.

6. All STEPS must be marked and not only the final answer. TICKS must correspond with the total.

7. If the marker comes across a novel or unusual solution please discuss it with the chief marker before proceeding.

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QUESTION 1

1.1	1.1.1 1.1.2 1.1.3 1.1.4 1.1.5	C A D A D	Ignore working (5 × 1)	 (5)

1.2	$\frac{(2-3i)-(3+2i)}{(3-2i)-(2-i)}$ $=\frac{(-1-5i)}{1-i}\times\frac{(1+i)}{(1+i)}$ ✓ $=\frac{-1-i-5i-5i^2}{1-i^2}$ $=\frac{4-6i}{2}=2-3i$	**Accuracy marks** ½ each for numerator and denominator. 1 for multiplying by conjugate ½ for real part, ½ for imaginary part.	 (3)

1.3.1	$z_1=12\ cis\ 120°$ $z_2=10\ cis\ 150°$	**Accuracy marks**	 (1)

1.3.2	z_1+z_2 $=12cis120°+10cis150$ $=-6+10,39i$ ✓ $-8,66+5i$ ✓ $=-14,66+15,39i$ ✓ *Alternative* $12(\cos120°+i\sin120°)+10(\cos150°+i\sin150°)$ $=12\left(-\frac{1}{2}+\frac{\sqrt{3}}{2}i\right)+10\left(\frac{-\sqrt{3}}{2}+\frac{1}{2}i\right)$ $=\left(-6+6\sqrt{3}i\right)$ ✓ $+\left(-5\sqrt{3}+5i\right)$ ✓ $=-6-5\sqrt{3}$ ✓$^{\frac{1}{2}}+\left(5+6\sqrt{3}\right)i$ ✓$^{\frac{1}{2}}$	1 for each conversion 1 for correct answer Allocation as indicated	 (3)

1.4	$\frac{\sqrt{18}\left(cis\,30°\times 3\right)}{\left(\sqrt{2}\right)^4 cis\left(45°\times 4\right)}$ $=\frac{\left(3\sqrt{2}\angle 90°\right)^{\frac{1}{2}}\checkmark}{\left(4\angle 180°\right)^{\frac{1}{2}}\checkmark}$ *or* $=\frac{4,243\angle 90°}{4\angle 180°}$ $=\frac{3\sqrt{2}}{4}\angle 90°-180°$ $=1,06\checkmark^{\frac{1}{2}}\angle -90°\checkmark^{\frac{1}{2}}$ *or* $\frac{3\sqrt{2}}{4}\angle -90°$ $=1,06\angle 270°\checkmark$ *or* $0,75\sqrt{2}\angle 270°$	½ each for applying De Moivre ½ for simplifying mod ½ for argument 1 for positive Argument	(3)
1.5	$(4+y)i^5-(8x+y+4)i^2=4(x^2+3i)$ $(4+y)i+(8x+y+4)=4x^2+12i$ ✓ $4+y=12$ … ✓½ (i) $8x+y+4=4x^2$ … ✓½ (ii) From (i): $4+y=12$ $y=8$ ✓ From (ii): $8x+y+4=4x^2$ Substitute $y=8$: $8x+8+4=4x^2$ $4x^2-8x-12=0$ $x^2-2x-3=0$ $(x+1)(x-3)=0$ $x=-1$ ✓; 3 ✓	1 for simplifying ½ for equation (i) ½ for equation (ii) 1 for correct y. 1 for each x value (CA)	(5)
			[20]

QUESTION 2

2.1.1	$f(x)=4x^3-\frac{5}{\sqrt[3]{x^2}}-\frac{a}{x}-\frac{\ln x}{4}-3^4$ $f(x)=4x^3-5(x)^{\frac{-2}{3}}\checkmark^{\frac{1}{2}}-ax^{-1}-\frac{1}{4}\ln x-3^4$ $f'(x)=12x^2\checkmark^{\frac{1}{2}}+\frac{10}{3}x^{\frac{-5}{3}}\checkmark^{\frac{1}{2}}+ax^{-2}\checkmark^{\frac{1}{2}}-\frac{1}{4x}\checkmark^{\frac{1}{2}}-0\checkmark^{\frac{1}{2}}$ $f'(x)=12x^2+\frac{10}{3x^{\frac{5}{3}}}\checkmark^{\frac{1}{2}}+\frac{a}{x^2}\checkmark^{\frac{1}{2}}-\frac{1}{4x}-0$ $f'(x)=12x^2+\frac{10}{3\sqrt[3]{x^5}}\checkmark^{\frac{1}{2}}+\frac{a}{x^2}-\frac{1}{4x}-0$	½ each for terms 2 **and 3** ½ for each derivative (A) ½ each for positive index (CA) ½ for changing to surd form	(5)
2.1.2	$y=\frac{e^{4x}}{\tan 4x}$ $\frac{dy}{dx}=\frac{vu'-uv'}{v^2}\checkmark^{\frac{1}{2}}$ $\frac{dy}{dx}=\frac{(\tan 4x)(4e^{4x})\checkmark^{\frac{1}{2}}\checkmark^{\frac{1}{2}}-(e^{4x})(4\sec^2 4x)\checkmark^{\frac{1}{2}}\checkmark^{\frac{1}{2}}}{(\tan 4x)^2\checkmark^{\frac{1}{2}}}$ $\frac{dy}{dx}=\frac{4e^{4x}(\tan 4x-\sec^2 4x)}{(\tan 4x)^2}$	½ apply quotient rule 1 for u' 1 for v' ½ for v^2	(3)

2.1.3	*Function of function*: $y=(3x^2-5)^5$ $\frac{dy}{dx}=5(3x^2-5)^4\checkmark(6x)\checkmark$ $\frac{dy}{dx}=(30x)(3x^2-5)^4\checkmark$ ✓ *differentiate exponent* ✓ *differentiate base* ✓ *simplify*	*Chain Rule*: *Let* $u=3x^2-5$ $\frac{du}{dx}=6x\checkmark^{\frac{1}{2}}$ $\therefore y=(u)^5$ $\therefore \frac{dy}{du}=5(u)^4\checkmark^{\frac{1}{2}}$ $\therefore \frac{dy}{dx}=\frac{dy}{du}\times\frac{du}{dx}$ $\therefore \frac{dy}{dx}=5(u)^4\checkmark^{\frac{1}{2}}(6x)\checkmark^{\frac{1}{2}}$ $\therefore \frac{dy}{dx}=5(3x^2-5)^4\checkmark^{\frac{1}{2}}(6x)$ $\frac{dy}{dx}=(30x)(3x^2-5)^4\checkmark^{\frac{1}{2}}$	$\checkmark^{\frac{1}{2}}\frac{du}{dx}$ $\checkmark^{\frac{1}{2}}\frac{dy}{du}$ ✓ *Substitute chain rule* $\checkmark^{\frac{1}{2}}$ *replace u* $\checkmark^{\frac{1}{2}}$ *simplify*	(3)

2.2.1	Distance at t = 2 $s=t^3\ln t=2^3\ln 2=5{,}545$ ✓	Accurate answer	(1)

2.2.2	Acceleration of the object. $s=t^3\ln t$ $\frac{ds}{dt}=t^3\left(\frac{1}{t}\right)+\ln t\,.3t^2$ ✓ $\frac{ds}{dt}=t^2+3t^2\ln t$ $\frac{d^2s}{dt^2}=2t+3t^2\left(\frac{1}{t}\right)+\ln t\,.6t$ ✓ $\frac{d^2s}{dt^2}=2t+3t+6t\,.\ln t$ $=5t+6t\ln t$ ✓	½ for derivative of $\ln t$ (A) ½ for derivative of t^3 (A) ½ each for **3 derivatives** (CA) ½ for simplification (CA)	(3)

2.3	Find $\frac{dy}{dx}$ from first principles. $\lim_{h\to 0}\frac{f(x+h)-f(x)}{h}$ $=\lim_{h\to 0}\frac{\left(3-2(x+h)\right)-(3-2x)}{h}$ ✓½ $=\lim_{h\to 0}\frac{(3-2x-2h)-3+2x}{h}$ ✓½ $=\lim_{h\to 0}\frac{-2h}{h}$ ✓½ $=-2$ ✓½ *no marks if steps are not shown*	½ for substitution (A) ½ for opening brackets (A) ½ for simplifying numerator (A) ½ for answer (A)	(2)

2.4.1	$f'(x)=12x^2+30x-18$ ✓ *At* $f'(x)=0$ *at TP* $\therefore 12x^2+30x-18=0$ ✓½ $6(2x^2+5x-3)=0$ $x=(x+3)(2x-1)$ ✓½ $x=-3$ ✓½ *or* $x=\frac{1}{2}$ ✓½	1 for derivative ½ setting derivative equal zero (CA) ½ for factors ½ each for 2 *x* coordinate of TP's	(3)

2.4.2	$\frac{d^2y}{dx^2}=24x+30$ ✓ $At\ \ x=-3:\ \ \frac{d^2y}{dx^2}=24(-3)+30<0$ Maximum at $x=-3$ ✓½ $At\ \ x=\frac{1}{2}:\ \ \frac{d^2y}{dx^2}=24\left(\frac{1}{2}\right)+30>0$ $\therefore Minimum\ \ at\ \ x=\frac{1}{2}$ ✓½	1 for derivative (CA) ½ for max point (CA) ½ for min point (CA)	(2)

2.4.3	$\frac{d^2y}{dx^2}=24x+30=0$ ✓½ $\therefore 24x=-30$ $\therefore x=\frac{-30}{24}=\frac{-5}{4}=-1,25$ ✓½ $y=4x^3+15x^2-18x+6$ $y=4(-1,25)^3+15(-1,25)^2-18(-1,25)+6$ ✓½ $y=\frac{353}{8}=44,125$ $POI=(-1,25\ ;44,125)$ ✓½	**Note: Ignore all ✓½ in the solution.** Instead, allocate: 1 for x coordinate of POI (CA) 1 for y coordinate of POI (CA)	(2)

2.4.4	Graph 2 ✓	1 for correct answer	(1)
			[25]

QUESTION 3

3.1.1	$\int(\frac{9}{4x}-\frac{12}{\sqrt{4x}}+e^{\frac{x}{2}}+4\sec^2 4x)\,dx$ $\int(\ \frac{9}{4x}-6x^{\frac{-1}{2}}\surd+e^{\frac{x}{2}}+4\sec^2 4x\ )\,dx$ $=\frac{9}{4}\ln x\checkmark^{\frac{1}{2}}-\frac{6x^{\frac{1}{2}}}{\frac{1}{2}}\checkmark^{\frac{1}{2}}+\frac{e^{\frac{x}{2}}}{\frac{1}{2}}\checkmark^{\frac{1}{2}}+\frac{4\tan 4x}{4}\checkmark^{\frac{1}{2}}+c\checkmark^{\frac{1}{2}}$ $=\frac{9}{4}\ln x-12x^{\frac{1}{2}}+2e^{\frac{x}{2}}+\tan 4x+c$ $=\frac{9}{4}\ln x-12\checkmark^{\frac{1}{2}}\sqrt{x}\checkmark^{\frac{1}{2}}+2e^{\frac{x}{2}}\checkmark^{\frac{1}{2}}+\tan 4x+c$ **Maximum of 3 marks if not simplified**	½ each for terms **two and three** ½ for each correct integration. (CA) ½ each for simplifying term 2, 3, 4 and ½ mark for surd (CA)	(5)
3.1.2	$\int(2x+\sin 3x+4x^3+e^{3x})\,dx$ $=\frac{2x^2}{2}\checkmark^{\frac{1}{2}}-\frac{\cos 3x}{3}\checkmark^{\frac{1}{2}}+\frac{4x^4}{4}\checkmark^{\frac{1}{2}}+\frac{e^{3x}}{3}\checkmark^{\frac{1}{2}}+c$ $=x^2-\frac{\cos 3x}{3}+x^4+\frac{e^{3x}}{3}+c$	½ for each correct integration	(2)
3.2	Distance: $\int_0^3(t^2-5t+12)\ dt\ \checkmark^{\frac{1}{2}}$ $=\frac{t^3}{3}\checkmark^{\frac{1}{2}}-\frac{5t^2}{2}\checkmark^{\frac{1}{2}}+12t\ \Big|_0^3\checkmark^{\frac{1}{2}}$ $=\left[\frac{(3)^3}{3}-\frac{5(3)^2}{2}+12(3)\right]-\left[\frac{0^3}{3}-\frac{5(0)^2}{2}+12(0)\right]$ $=22,5\ m\checkmark$ *do not penalise if SI – unit is omited*	½ for substitution ½ for each integral 1 for correct answer	(3)

3.3	$A=\int_a^b y\,dx$ $A_1=\int_0^3\left(x^3-6x^2+9x+2\right)dx$ ✓½✓½ $=\frac{x^4}{4}-2x^3+\frac{9}{2}x^2+2x\ \Big\|_0^3$ ✓½✓½✓½✓½ $=\left[\frac{(3)^4}{4}-2(3)^3+\frac{9}{2}(3)^2+2(3)\right]$✓½$-\left[\frac{(0)^4}{4}-2(0)^3+\frac{9}{2}(0)^2+2(0)\right]$✓½ $=12,75km^2$✓	1 for correct substitution ½ per integral ½ and ½ for substitution (CA) 1 for correct answer	(5)
			[15]

QUESTION 4

4.1	$\left(2x^3-x^2-px+20\right)\div(x+4)$ Re $mainder=4$ $f(-4)=2(-4)^3-(-4)^2-p(-4)+20$ ✓½$=4$ $\therefore -124+4p=4$ ✓½ $\therefore 4p=128$ $\therefore p=32$✓	½ for substituting $x=4$ ½ for equation in p 1 for answer	(2)

4.2	Elimination $f(x)=2x^3-x^2-ax+b$ $At\ x=\frac{1}{2}\ f(x)=0$ $\therefore f(\frac{1}{2})=2\left(\frac{1}{2}\right)^3-\left(\frac{1}{2}\right)^2-a\left(\frac{1}{2}\right)+b=0$ ✓½ $\therefore -\frac{a}{2}+b=0 \qquad eq.1$✓ $(2;-36)\ point\ on\ f(x)=2x^3-x^2-ax+b$ $\therefore -36=2(2)^3-(2)^2-a(2)+b$ $\therefore -2a+b=-48 \qquad eq.2$✓ $Elimination:$ $eq1-eq2:\ \left(-\frac{a}{2}+b\right)-(-2a+b)=0-(-48)$ ✓½ $\frac{3}{2}a=48$ $\therefore a=32$✓ $Substitute\ in\ eq1$ $\therefore b=\frac{32}{2}=16$✓	Substitution $f(x)=2x^3-x^2-ax+b$ $At\ x=\frac{1}{2}\ f(x)=0$ $\therefore f(\frac{1}{2})=2\left(\frac{1}{2}\right)^3-\left(\frac{1}{2}\right)^2-a\left(\frac{1}{2}\right)+b=0$ ✓½ $\therefore -\frac{a}{2}+b=0 \qquad eq.1$✓ $(2;-36)\ point\ on\ f(x)=2x^3-x^2-ax+b$ $\therefore -36=2(2)^3-(2)^2-a(2)+b$ $\therefore -2a+b=-48 \qquad eq.2$✓ $Substitution\ in\ eq2:\ b=\frac{a}{2}\ from\ eq1$ $\therefore\ -2a+\frac{a}{2}=-48$ ✓½ $\therefore\ \frac{3}{2}a=48$ $\therefore a=32$✓ $Substitute\ in\ eq1$ $\therefore b=\frac{32}{2}=16$✓	(5)

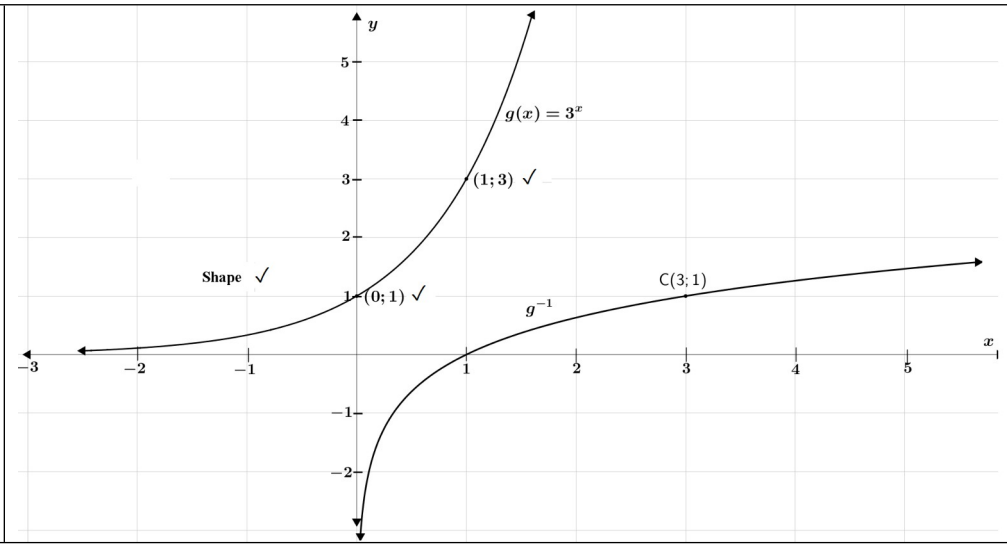
4.3.1	From graph: $m=\frac{vertical}{horizontal}=\frac{2}{3}$ ✓ $c=y\ intercept=-2$ **Alternate:** $(3,0)$ and $(0;-2)$ are points on h. $m=\frac{y_2-y_1}{x_2-x_1}=\frac{0-(-2)}{3-0}=\frac{2}{3}$ $c=y\ intercept=-2$	1 for correct m value 1 for correct c value 1 for correct m value 1 for correct c value	(2)

4.3.2	Domain of h: $-1\le x<6$ When $x=-1$, $y=\frac{2}{3}(-1)-2=-\frac{8}{3}$ ✓½ When $x=6$, $y=\frac{2}{3}(6)-2=2$ ✓½	½ for lower limit ½ for upper limit 1 for correct inequality	(2)

	Range of h: $\frac{-8}{3} \leq y < 2$ ✓		

4.3.3	$C(3;1)$ lies on g^{-1} Hence $(1;3)$ lies on g From $g(x)=a^x$, $3=a^1$ $\therefore g(x)=3^x$	½ for point on g ½ for answer **Full marks for answer only**	(1)

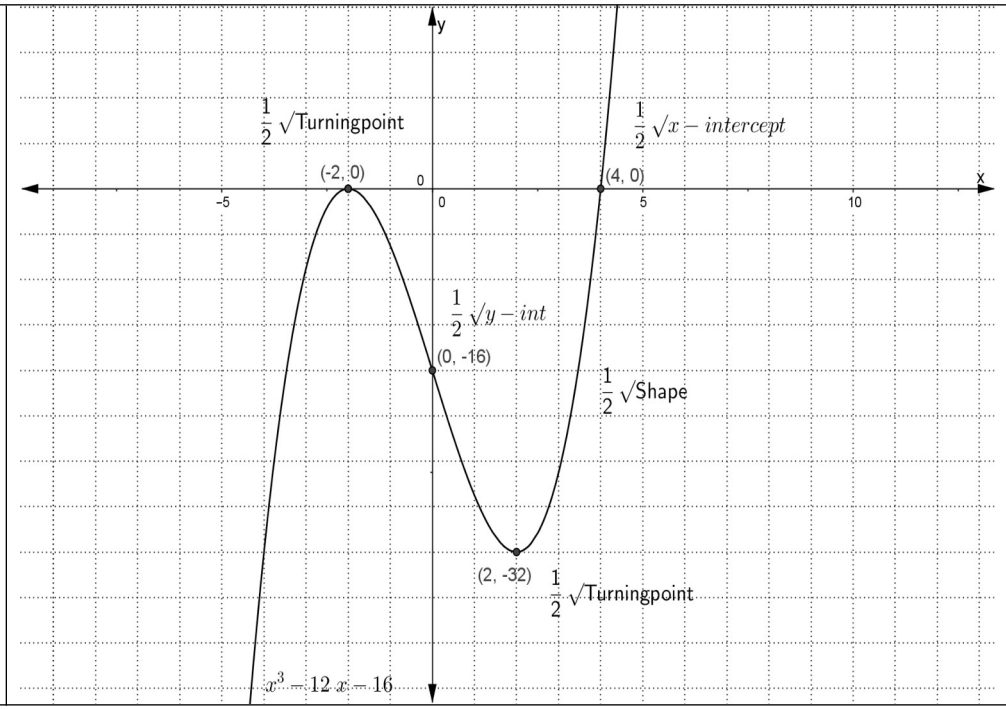
4.3.4	Equation of g^{-1}: $x=3^y$	1 for answer ✓ (CA)	(1)

4.3.5		1 each for the two points. 1 for shape	(3)

4.3.6	Line of symmetry: $y=x$ ✓	1 for correct answer	(1)

4.4.1	For y-*intercept ; let* $x=0$ $\therefore y=0^3-12(0)-16=-16$ ✓ Factor theorem: $f(-2)=(-2)^3-12(-2)-16=0$ ✓ $\therefore x+2$ is a factor $f(4)=(4)^3-12(4)-16=0$ ✓ $\therefore x-4$ is factor $\therefore (x+2)(x-4)(x+a)=x^3-12x-16$ By inspection, $2\times-4\times a=-16$	1 for correct y intercept 1 for any first zero 1 for second or third zero obtained any correct method.	(5)

	$\therefore a = 2$ $x\text{-}intercepts: x\text{-}2$ ✓ ; $=4$ ✓	1 for each correct x intercept	

4.4.2		½ each for x intercepts ½ for y intercept 1 for min turning point **(not ½)** ½ for shape	(3)
			[25]

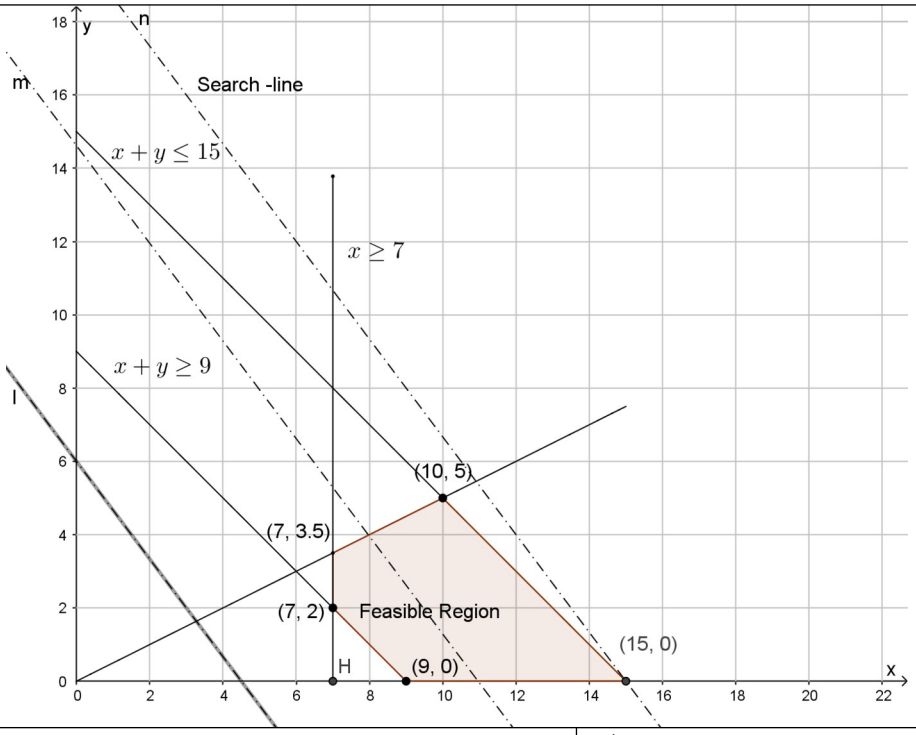
QUESTION 5

5.1.1	$y \leq 2x + 2$ ✓ $\frac{x}{3} + \frac{y}{6} < 1$ ✓ $y \geq -x + \frac{7}{2}$ ✓	1 for each correct inequality	(3)

5.1.2	No. ✓½ Point (2; 2) lies on the line $\frac{x}{3} + \frac{y}{6} = 1$ which is not in the feasible region. ✓½	½ for correct answer ½ for reason	(1)

5.2.1	$x+y\geq 9$ ✓ $x+y\leq 15$ ✓½ $\frac{x}{y}\geq\frac{2}{1}$ or $y\leq\frac{1}{2}x$ ✓ $x\geq 7$ ✓½	1 mark for 1st constraint ½ mark for 2nd constraint 1 mark for this constraint ½ mark for this constraint	(3)

5.2.2	$C=80x+60y$ ✓	1 for correct answer	(1)

5.2.3		1 for each correctly drawn line: $x+y\geq 9$ $x+y\leq 15$ $y\leq\frac{1}{2}x$ $x\geq 7$ 1 for shading feasible region (CA) marks	
	✓ ✓ ✓ ✓ for correct boundary lines	✓ correct feasible region	(5)

5.2.4	Search line: $C=80x+60y$ ✓ For most costly mixture, x = 15 kg of beef ✓½ y = 0 kg of pork ✓½	1 mark on graph for one of dotted lines with gradient $\frac{-4}{3}$. ½ for x value (CA) ½ for y value (CA)	(2)
			[15]

	TOTAL: 100

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